Homework 05 - Law and courts

INFO 5001 - Fall 2025

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Setup

Load packages and data:

```r
install.packages(c("tidyverse", "rvest", "janitor"))
```

```
The following package(s) will be installed:
- janitor   [2.2.1]
- rvest     [1.0.5]
- tidyverse [2.0.0]
These packages will be installed into "~/.local/hw-05-jl4495-1/renv/library/
R-4.3/x86_64-pc-linux-gnu".

# Installing packages --------------------------------------------------------
- Installing rvest ...                          OK [linked from cache]
- Installing tidyverse ...                      OK [linked from cache]
- Installing janitor ...                        OK [linked from cache]
Successfully installed 3 packages in 39 milliseconds.
```

```r
library(tidyverse)
```

```
── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0
──
✔ dplyr     1.1.4     ✔ readr     2.1.5
✔ forcats   1.0.1     ✔ stringr   1.5.2
✔ ggplot2   4.0.0     ✔ tibble    3.3.0
✔ lubridate 1.9.4     ✔ tidyr     1.3.1
✔ purrr     1.1.0
── Conflicts ────────────────────────────────────────── tidyverse_conflicts()
──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
```

```
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
conflicts to become errors
```

```r
library(scales)
```

```


Attaching package: 'scales'

The following object is masked from 'package:purrr':

    discard

The following object is masked from 'package:readr':

    col_factor
```

```r
library(rvest)
```

```


Attaching package: 'rvest'

The following object is masked from 'package:readr':

    guess_encoding
```

Exercise 1

```r
library(rvest)
library(tidyverse)
library(janitor)
```

```


Attaching package: 'janitor'
```

```
The following objects are masked from 'package:stats':

    chisq.test, fisher.test
```

```r
url <- "https://www.prisonpolicy.org/phones/appendices2022_1.html"
tables <- read_html(url) |> html_table(fill = TRUE)
phone_raw <- tables[[1]]
```

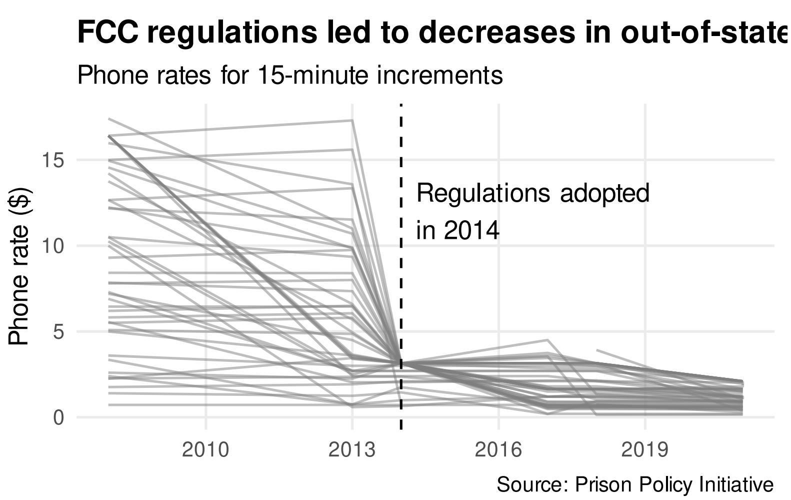
```r
header1 <- phone_raw[1, ]
header2 <- phone_raw[2, ]
new_header <- ifelse(header2 == "", header1, paste(header1, header2, sep =
"_"))
colnames(phone_raw) <- new_header

phone_raw <- phone_raw[-c(1, 2), ]
phone_raw <- janitor::clean_names(phone_raw)
out_state_raw <- phone_raw[, c(1, 9:ncol(phone_raw))]

out_state_tidy <- out_state_raw |>
  rename(states = 1) |>
  pivot_longer(
    cols = -states,
    names_to = "year",
    values_to = "rate"
  ) |>
  mutate(
    type = "out_state",
    year = readr::parse_number(year),
    rate = readr::parse_number(rate)
  )
```

```
Warning: There were 2 warnings in `mutate()`.
The first warning was:
ℹ In argument: `year = readr::parse_number(year)`.
Caused by warning:
! 49 parsing failures.
row col expected actual
  2  -- a number     na
 10  -- a number     na
 18  -- a number     na
 26  -- a number     na
 34  -- a number     na
... ... ........ ......
See problems(...) for more details.
ℹ Run `dplyr::last_dplyr_warnings()` to see the 1 remaining warning.
```

```r
ggplot(out_state_tidy, aes(x = year, y = rate, group = states)) +
  geom_line(alpha = 0.5, color = "gray50") +
  geom_vline(xintercept = 2014, linetype = "dashed") +
  annotate(
    "text",
    x = 2014.3,
    y = 12,
```

```
    label = "Regulations adopted\nin 2014",
    hjust = 0
  ) +
  labs(
    title = "FCC regulations led to decreases in out-of-state prison phone
rates",
    subtitle = "Phone rates for 15-minute increments",
    x = NULL,
    y = "Phone rate ($)",
    caption = "Source: Prison Policy Initiative"
  ) +
  theme_minimal(base_size = 13) +
  theme(
    plot.title = element_text(face = "bold"),
    panel.grid.minor = element_blank()
  )
```

```
Warning: Removed 51 rows containing missing values or values outside the scale
range
(`geom_line()`).
```



Exercise 2

```
scdb_case <- read_csv("data/scdb-case.csv")
```

```
Rows: 29227 Columns: 52
── Column specification
────────────────────────────────────────────────────────
Delimiter: ","
```

```
chr (14): caseId, docketId, caseIssuesId, dateDecision, usCite, sctCite,
led...
dbl (38): decisionType, term, naturalCourt, petitioner, petitionerState,
res...

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this
message.
```

```
scdb_vote <- read_csv("data/scdb-vote.csv")
```

```
Rows: 255857 Columns: 13
── Column specification
────────────────────────────────────────────────────────
Delimiter: ","
chr (5): caseId, docketId, caseIssuesId, voteId, justiceName
dbl (8): term, justice, vote, opinion, direction, majority,
firstAgreement, ...

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this
message.
```

```
scdb_joined <- scdb_case |>
  left_join(scdb_vote, by = "caseId")
glimpse(scdb_joined)
```

```
Rows: 255,857
Columns: 64
$ caseId                     <chr> "1791-001", "1791-001", "1791-001",
"1791-001…
$ docketId.x                 <chr> "1791-001-01", "1791-001-01", "1791-001-01",
…
$ caseIssuesId.x             <chr> "1791-001-01-01", "1791-001-01-01",
"1791-001…
$ dateDecision               <chr> "8/3/1791", "8/3/1791", "8/3/1791",
"8/3/1791…
$ decisionType               <dbl> 6, 6, 6, 6, 6, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2,
…
$ usCite                     <chr> "2 U.S. 401", "2 U.S. 401", "2 U.S. 401", "2
…
$ sctCite                    <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
N…
$ ledCite                    <chr> "1 L. Ed. 433", "1 L. Ed. 433", "1 L. Ed.
```

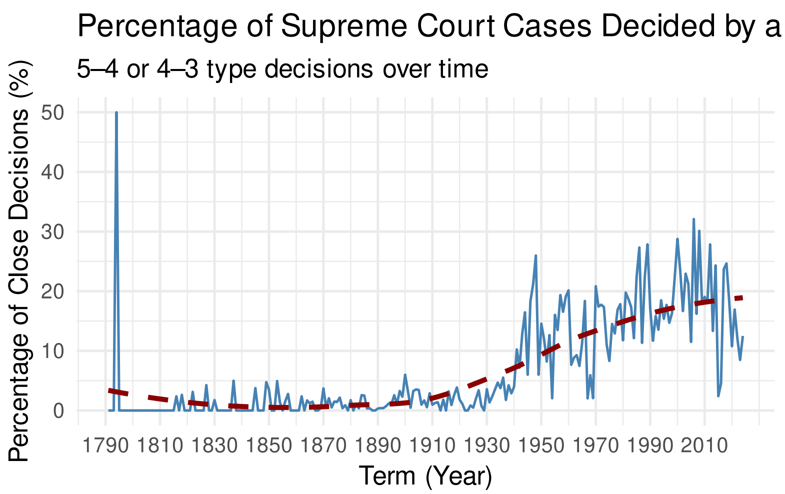
```
433…
$ lexisCite                <chr> "1791 U.S. LEXIS 189", "1791 U.S. LEXIS
189",…
$ term.x                   <dbl> 1791, 1791, 1791, 1791, 1791, 1791, 1791,
179…
$ naturalCourt             <dbl> 102, 102, 102, 102, 102, 102, 102, 102, 102,
…
$ chief                    <chr> "Jay", "Jay", "Jay", "Jay", "Jay", "Jay",
"Ja…
$ docket                   <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
N…
$ caseName                 <chr> "WEST, PLS. IN ERR. VERSUS BARNES. et al.",
"…
$ dateArgument             <chr> "8/2/1791", "8/2/1791", "8/2/1791",
"8/2/1791…
$ dateRearg                <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
N…
$ petitioner               <dbl> 501, 501, 501, 501, 501, 501, 501, 501, 501,
…
$ petitionerState          <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
N…
$ respondent               <dbl> 111, 111, 111, 111, 111, 28, 28, 28, 28, 28,
…
$ respondentState          <dbl> NA, NA, NA, NA, NA, 25, 25, 25, 25, 25, 37,
3…
$ jurisdiction             <dbl> 13, 13, 13, 13, 13, 15, 15, 15, 15, 15, 15,
1…
$ adminAction              <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
N…
$ adminActionState         <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
N…
$ threeJudgeFdc            <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
…
$ caseOrigin               <dbl> 431, 431, 431, 431, 431, NA, NA, NA, NA, NA,
…
$ caseOriginState          <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
N…
$ caseSource               <dbl> 431, 431, 431, 431, 431, NA, NA, NA, NA, NA,
…
$ caseSourceState          <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
N…
$ lcDisagreement           <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
…
$ certReason               <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
…
$ lcDisposition            <dbl> NA, NA, NA, NA, NA, 9, 9, 9, 9, 9, NA, NA,
NA…
$ lcDispositionDirection   <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
```

```
…
$ declarationUncon         <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
…
$ caseDisposition          <dbl> 9, 9, 9, 9, 9, 1, 1, 1, 1, 1, 9, 9, 9, 9, 9,
…
$ caseDispositionUnusual   <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
…
$ partyWinning             <dbl> 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, 0, 0, 0, 0, 0,
…
$ precedentAlteration      <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
…
$ voteUnclear              <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
…
$ issue                    <dbl> 90320, 90320, 90320, 90320, 90320, 90320,
903…
$ issueArea                <dbl> 9, 9, 9, 9, 9, 9, 9, 9, 9, 9, 9, 9, 9, 9, 9,
…
$ decisionDirection        <dbl> 1, 1, 1, 1, 1, 2, 2, 2, 2, 2, 1, 1, 1, 1, 1,
…
$ decisionDirectionDissent <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
…
$ authorityDecision1       <dbl> 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3,
…
$ authorityDecision2       <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
N…
$ lawType                  <dbl> 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 6, 6, 6, 6, 6,
…
$ lawSupp                  <dbl> 403, 403, 403, 403, 403, 403, 403, 403, 403,
…
$ lawMinor                 <chr> "NULL", "NULL", "NULL", "NULL", "NULL",
"NULL…
$ majOpinWriter            <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
N…
$ majOpinAssigner          <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
…
$ splitVote                <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
…
$ majVotes                 <dbl> 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 6, 6, 6, 6, 6,
…
$ minVotes                 <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
…
$ docketId.y               <chr> "1791-001-01", "1791-001-01", "1791-001-01",
…
$ caseIssuesId.y           <chr> "1791-001-01-01", "1791-001-01-01",
"1791-001…
$ voteId                   <chr> "1791-001-01-01-01-01",
"1791-001-01-01-01-02…
$ term.y                   <dbl> 1791, 1791, 1791, 1791, 1791, 1791, 1791,
```

```
179…
$ justice                  <dbl> 1, 3, 4, 5, 6, 1, 3, 4, 5, 6, 1, 3, 4, 5, 6,
…
$ justiceName              <chr> "JJay", "WCushing", "JWilson", "JBlair",
"JIr…
$ vote                     <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
…
$ opinion                  <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
…
$ direction                <dbl> 1, 1, 1, 1, 1, 2, 2, 2, 2, 2, 1, 1, 1, 1, 1,
…
$ majority                 <dbl> 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2,
…
$ firstAgreement           <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
N…
$ secondAgreement          <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
N…
```

```
one_vote_margin <- scdb_joined |>
  filter(!is.na(majVotes), !is.na(minVotes)) |>
  mutate(one_vote_margin = abs(majVotes - minVotes) == 1)

one_vote_summary <- one_vote_margin |>
  group_by(term.y) |>
  summarise(
    total_cases = n(),
    one_vote_margin_cases = sum(one_vote_margin, na.rm = TRUE),
    pct_one_vote = (one_vote_margin_cases / total_cases) * 100
  )

  ggplot(one_vote_summary, aes(x = term.y, y = pct_one_vote)) +
  geom_line(color = "steelblue") +
  geom_smooth(method = "loess", se = FALSE, color = "darkred", linetype =
"dashed") +
  labs(
    title = "Percentage of Supreme Court Cases Decided by a One-Vote Margin",
    subtitle = "5–4 or 4–3 type decisions over time",
    x = "Term (Year)",
    y = "Percentage of Close Decisions (%)"
  ) +
  scale_x_continuous(breaks = seq(1790, 2020, by = 20)) + # makes term axis
continuous
  theme_minimal(base_size = 13)
```

```
`geom_smooth()` using formula = 'y ~ x'
```

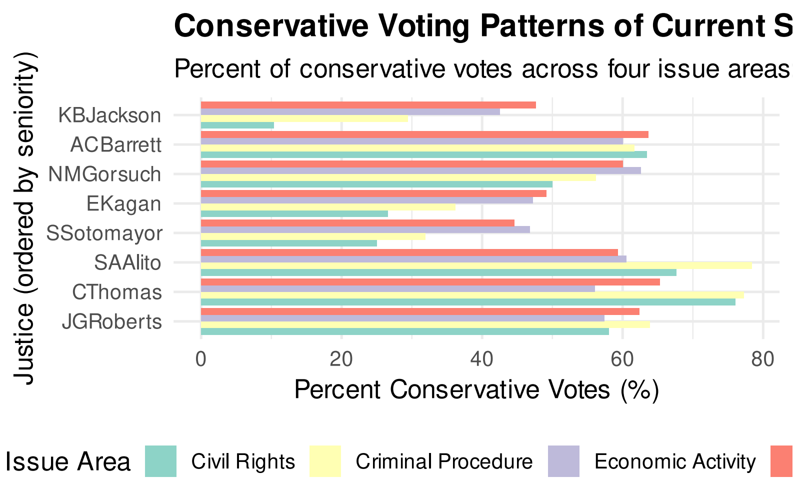



```
# According to the graph, the percentage of on-vote margin starts to gradually
increase from the starting from 1895.
```

Exercise 3

```
scdb_joined <- scdb_joined |>
  mutate(
    issueAreaLabel = recode(as.character(issueArea),
      "1" = "Criminal Procedure",
      "2" = "Civil Rights",
      "8" = "Economic Activity",
      "9" = "Judicial Power",
      .default = NA_character_
    )
  )

scdb_filtered <- scdb_joined |>
  filter(!is.na(issueAreaLabel), !is.na(direction))

scdb_filtered <- scdb_filtered |>
  mutate(conservative_vote = if_else(direction == 1, 1, 0))

current_justices <- tibble(
  justiceName = c(
    "JGRoberts",
    "CThomas",
    "SAAlito",
    "SSotomayor",
    "EKagan",
    "NMGorsuch",
    "BKavanaugh",
```

```r
    "ACBarrett",
    "KBJackson"
  ),
  seniority_rank = 1:9
)

conservative_summary <- scdb_filtered |>
  filter(justiceName %in% current_justices$justiceName) |>
  group_by(justiceName, issueAreaLabel) |>
  summarise(
    total_votes = n(),
    conservative_votes = sum(conservative_vote, na.rm = TRUE),
    pct_conservative = (conservative_votes / total_votes) * 100,
    .groups = "drop"
  ) |>
  left_join(current_justices, by = "justiceName")

ggplot(conservative_summary, aes(
  x = reorder(justiceName, seniority_rank),
  y = pct_conservative,
  fill = issueAreaLabel
)) +
  geom_col(position = "dodge") +
  coord_flip() +
  scale_fill_brewer(palette = "Set3") +
  labs(
    title = "Conservative Voting Patterns of Current Supreme Court Justices",
    subtitle = "Percent of conservative votes across four issue areas:
Criminal Procedure, Civil Rights, Economic Activity, and Judicial Power",
    x = "Justice (ordered by seniority)",
    y = "Percent Conservative Votes (%)",
    fill = "Issue Area"
  ) +
  theme_minimal(base_size = 13) +
  theme(
    plot.title = element_text(face = "bold"),
    legend.position = "bottom"
  )
```




```
#
```

Exercise 4

```r
library(tidyverse)
library(lubridate)

scdb_case <- read_csv("data/scdb-case.csv")
```

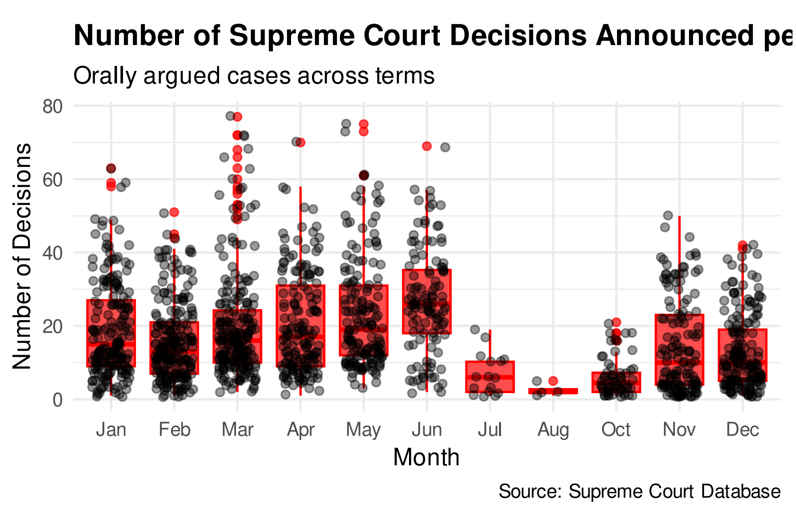
```
Rows: 29227 Columns: 52
── Column specification
────────────────────────────────────────────────────────
Delimiter: ","
chr (14): caseId, docketId, caseIssuesId, dateDecision, usCite, sctCite,
led...
dbl (38): decisionType, term, naturalCourt, petitioner, petitionerState,
res...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this
message.
```

```r
scdb_oral <- scdb_case |>
  filter(decisionType == 1, !is.na(dateDecision)) |>
  mutate(
    dateDecision = mdy(dateDecision)
  )

scdb_oral <- scdb_oral |>
```

```
  mutate(
    month = month(dateDecision, label = TRUE, abbr = TRUE),
    month_num = month(dateDecision)
  )

decisions_by_month <- scdb_oral |>
  group_by(term, month_num, month) |>
  summarise(
    n_decisions = n(),
    .groups = "drop"
  )

nrow(decisions_by_month)
```

```
[1] 1478
```

```
ggplot(decisions_by_month, aes(x = month, y = n_decisions)) +
  geom_boxplot(alpha = 0.7, color = "red", fill = "red")+
  geom_jitter(width = 0.3, alpha = 0.4) +
  labs(
    title = "Number of Supreme Court Decisions Announced per Month",
    subtitle = "Orally argued cases across terms",
    x = "Month",
    y = "Number of Decisions",
    caption = "Source: Supreme Court Database"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    plot.title = element_text(face = "bold")
  )
```



```
  #  Conservative justices like Thomas, Alito, and Roberts vote conservatively
most often, while justices like Sotomayor, Kagan, and Jackson do so far less.
Barrett and Gorsuch fall in between, showing variation across issue types.
Overall, this highlights a clear ideological divide on the Court.
```

Exercise 5

```r
library(tidyverse)

vote <- read_csv("data/scdb-vote.csv") |> rename_all(tolower)
```

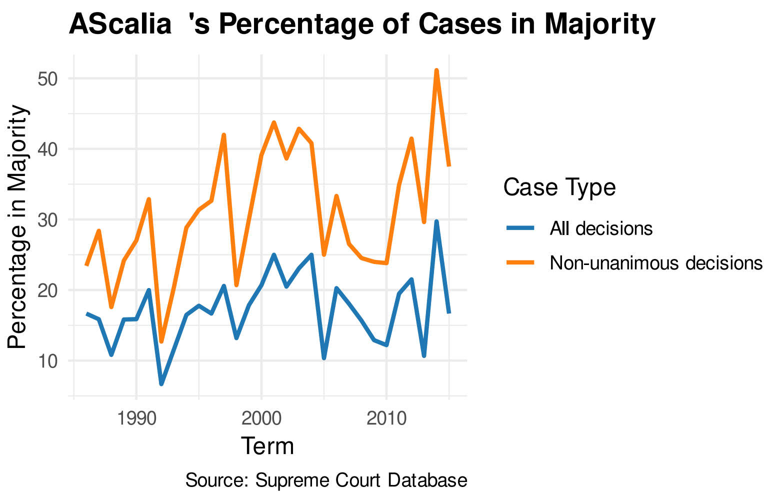
```
Rows: 255857 Columns: 13
── Column specification
────────────────────────────────────────────────────────
Delimiter: ","
chr (5): caseId, docketId, caseIssuesId, voteId, justiceName
dbl (8): term, justice, vote, opinion, direction, majority,
firstAgreement, ...

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this
message.
```

```r
case <- read_csv("data/scdb-case.csv") |> rename_all(tolower)
```

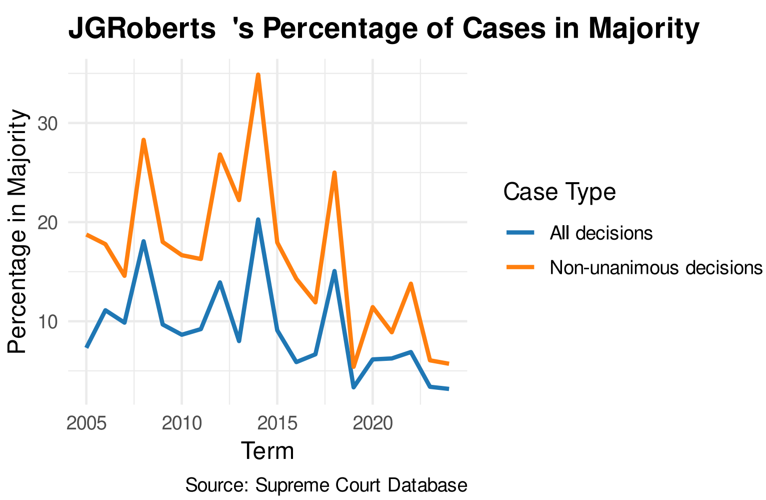
```
Rows: 29227 Columns: 52
── Column specification
────────────────────────────────────────────────────────
Delimiter: ","
chr (14): caseId, docketId, caseIssuesId, dateDecision, usCite, sctCite,
led...
dbl (38): decisionType, term, naturalCourt, petitioner, petitionerState,
res...

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this
message.
```

```r
votes_joined <- vote |>
  left_join(
    case |>
      select(caseid, term, majvotes, minvotes, decisiontype),
    by = "caseid"
  ) |>
```

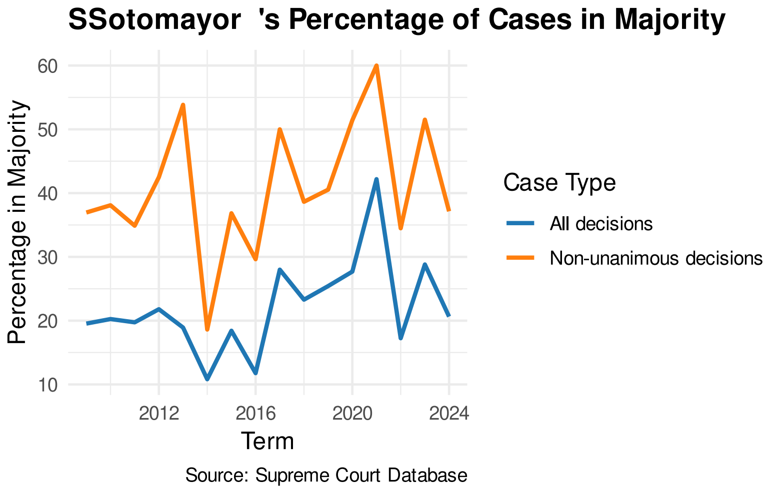
```r
  mutate(
    in_majority = if_else(majority == 1, 1, 0),
    njustices = majvotes + minvotes,
    non_unanimous = if_else(majvotes != njustices, 1, 0)
  ) |>
  rename(term = term.x)

plot_majority_pct <- function(justice_name) {
  justice_stats <- votes_joined |>
    filter(justicename == justice_name) |>
    group_by(term) |>
    summarize(
      pct_majority_all = mean(in_majority, na.rm = TRUE) * 100,
      pct_majority_nonunanimous = mean(in_majority[non_unanimous == 1], na.rm
= TRUE) * 100,
      .groups = "drop"
    )

  ggplot(justice_stats, aes(x = term)) +
    geom_line(aes(y = pct_majority_all, color = "All decisions"), linewidth =
1) +
    geom_line(aes(y = pct_majority_nonunanimous, color = "Non-unanimous
decisions"), linewidth = 1) +
    scale_color_manual(
      values = c(
       "All decisions" = "#1F77B4",
"Non-unanimous decisions" = "#FF7F0E"

      )
    ) +
    labs(
      title = paste(justice_name, " 's Percentage of Cases in Majority"),
      x = "Term",
      y = "Percentage in Majority",
      color = "Case Type",
      caption = "Source: Supreme Court Database"
    ) +
    theme_minimal(base_size = 12) +
    theme(plot.title = element_text(face = "bold"))
}

plot_majority_pct("AScalia")    # Antonin Scalia
```




```r
plot_majority_pct("JGRoberts")  # John Roberts
```




```r
plot_majority_pct("SSotomayor") # Sonia Sotomayor
```




```
# all three justices have higher percentage of votes in non-unaimous
decisions.They are more likely to be aligned with the winning side in
contested cases (where there is dissent) than in all cases overall.
```